

Zhuohang Wu

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EDUCATION

University of California - Irvine

Irvine, CA

Master of Computer Science, GPA: 3.95

Sep 2025 - Dec 2026 (Expected)

Tongji University

Shanghai, China

Bachelor of Engineering in Software Engineering, GPA: 89.3/100

Sep 2021 - Jun 2025

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java,

Web Application & Databases: React.js, Node.js, Vue.js, SpringBoot, MySQL, Clickhouse, PostgreSQL

Cloud, Others: GCP, AWS, Apache Kafka, Kubernetes, Docker, PyTorch

RESEARCH EXPERIENCE

Shang Data Lab, UC San Diego

San Diego, United States

Research Intern

Feb 2026 - Present

- Proposed the **Adapted Codified Decision Tree (Adapted CDT)** framework that combines the interpretability of symbolic decision trees with the semantic reasoning of large language models, modeling group-level decision behaviors for **38 real-world entities across 15 industries**.
- Developed the Adapter Tree temporal adaptation algorithm: DP-optimal **phase partitioning** → **KMeans** dual-embedding clustering (less than 6 clusters) → sequential per-cluster GPT-4o semantic **refinement**, adapting decision logic to evolving behaviors while preserving tree structure.
- Constructed benchmark datasets of **10,000+** scene-action pairs from Wikipedia and TechCrunch; CDT methods achieved 66.5 NLI score, outperforming Profile/ETA/RICL baselines by **7-8 points**.
- Built a cross-organization CDT transfer framework via systematic entity substitution; quantified temporal behavior drift across 38 organizations (**mean drift +5.5%**, 22/38 significant at $p < 0.05$).

NaMI Lab, Tongji University

Shanghai, China

Research Intern

Dec 2024 - Sep 2025

- Researched on using reinforcement learning to solve **low-altitude last-mile logistics**. Used **K-means** clustering to route the truck and decouple collaboration among **heterogeneous agents** (truck, drone). Modeled into **Markov Decision Process (MDP)** and designed a **Transformer-based model** as drone's policy network and used **Proximal Policy Optimization (PPO)** algorithm to train the model.
- Numerical study showed that our approach improved **27.07%** in delivery time and **6.87%** in cost.

TACO-Group, Texas T&M University

College Station, United States

Research Intern

Mar 2025 - Aug 2025

- Proposed **SAFEFLOW**, a protocol-level framework for secure **LLM/VLM-based autonomous agents**, featuring fine-grained information flow control (IFC) with dynamic trust labeling to enforce runtime confidentiality and integrity across agents, tools, and environments.
- Designed transactional execution with write-ahead logging, DAG-based failure isolation, and rollback mechanisms for verifiable and recoverable agent operations. Developed concurrency control with task-aware scheduling to enable safe multi-agent coordination over shared resources.
- Built **SAFEFLOWBENCH**, the first benchmark for VLM-agent safety in GUI environments, comprising **332 multimodal threat scenarios** (MTST) and 25 multi-agent concurrency scenarios (CART).
- SAFEFLOW**-equipped agents achieved **94–99%** task completion with **0%** unsafe actions across 10 frontier VLMs, versus **56–79%** unsafe rates without the framework. Achieved **0%** harm score on all 14 models in cross-benchmark evaluation on AgentHarm.

WORK EXPERIENCE

Shanghai Wenbo Investment Management Co., Ltd.

Shanghai, China

Software Engineer Intern

Jul 2024 - Nov 2024

- Developed a **Kindle1m data pipeline**, managing over 2000 instruments from various exchanges; used **WebSocket** to obtain and send data to an **Apache Kafka** cluster with a single message delay under 0.1 seconds and a total delay under 1 second.
- Stored the data on **Clickhouse**; employed efficient **aggregation and deduplication** through MergeTree Family and aggregation functions. Developed a multi-processes Python program to monitor the pipeline and deployed it on **Kubernetes**; stored the results in Clickhouse and visualized them on **Grafana**.

PROJECTS

Dispatch+: Dispatch & Delivery Management App

- Developed a dispatch & delivery management app with **React.js, MUI, Flask and MySQL**.
- Developed **CRUD REST APIs** using Flask decorator, encompassing functionalities such as registration, ordering, checkout, monitoring and personal information management, and auth'ed users using **JWT**.
- Used **Google Map API** to generate routes, and deployed using docker to **Google Cloud Platform**.